# Official Component Requisition: HydroTrack v3.0 (Smart City Node)

## 1. Compute & Connectivity

| Item | Description | Qty | Unit Cost (INR) | Total Cost (INR) | Buying Link / Source |
|---|---|---|---|---|---|
| **Espressif ESP32-S3-DevKitC-1** | Official Espressif Dual-core Edge AI board (2 for nodes, 1 for gateway, 1 spare) | 4 | ₹1,539 | ₹6,156 | Robu.in - ESP32-S3 DevKit |

## 2. Sensor Suite & AI Diagnostics

| Item | Description | Qty | Unit Cost (INR) | Total Cost (INR) | Buying Link / Source |
|---|---|---|---|---|---|
| **Water Flow Sensor** | YF-S201C or equivalent in-line volumetric hall/vortex sensor | 3 | ₹300 | ₹900 | Robu.in - Water Flow Sensor |
| **DFRobot Analog TDS Sensor** | Official DFRobot water quality sensor with AC excitation + waterproof probes | 3 | ₹1,429 | ₹4,287 | Robu.in - DFRobot TDS |
| **27mm Piezoelectric Disc** | Contact microphone for TinyML acoustic leak/cavitation detection | 5 | ₹20 | ₹100 | Robu.in - 27mm Piezo |
| **10K NTC Thermistor** | Waterproof probe for | 3 | ₹100 | ₹300 | Robu.in - 10K NTC |

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| | temperature & viscosity math compensation | | | | Waterproof |

## 3. Deep-Sleep Power Architecture

| Item | Description | Qty | Unit Cost (INR) | Total Cost (INR) | Buying Link / Source |
|---|---|---|---|---|---|
| **TPS62840DLCR DC-DC IC** | Ultra-low 60nA quiescent current buck converter (Must have spares for soldering) | 5 | ₹215 | ₹1,075 | Mouser.in - TPS62840 |
| **800mAh LiPo Pouch Cell** | 3.7V Lithium-Polymer battery (compact size to fit inside 60mm tube) | 3 | ₹300 | ₹900 | Robu.in - 800mAh LiPo |
| **TP4056 Type-C Charger** | Battery charging module with protection circuit | 4 | ₹40 | ₹160 | Robu.in - TP4056 Type-C |

## 4. Bio-Health & Optics (The UI)

| Item | Description | Qty | Unit Cost (INR) | Total Cost (INR) | Buying Link / Source |
|---|---|---|---|---|---|
| **WS2812B RGB LED Ring** | 16-Bit individually addressable LED halo (outer diameter ~40mm) | 4 | ₹150 | ₹600 | Robu.in - WS2812B Ring 16 |
| **275nm UV-C SMD LED** | High-intensity Deep UV-C | 4 | ₹250 | ₹1,000 | IndiaMART - UV-C 275nm |

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| | diodes (3535 package) for sterilization | | | | |
| **Neoperl M22 Laminar Insert** | Creates splash-free, fiber-optic clear water stream for LEDs | 2 | ₹250 | ₹500 | Amazon.in - Neoperl M22 |

## 5. Fabrication, Waterproofing & Shielding

| Item | Description | Qty | Unit Cost (INR) | Total Cost (INR) | Buying Link / Source |
|---|---|---|---|---|---|
| **MG Chemicals 422B** | Silicone Conformal Coating (Crucial to waterproof the raw PCBs) | 1 | ₹3,850 | ₹3,850 | RS Components India - 422B |
| **Clear Epoxy Resin Kit** | Art-grade clear resin/hardener to seal and frost the WS2812B LED ring | 1 | ₹400 | ₹400 | Amazon.in - Clear Epoxy Resin |
| **Aluminum Foil HVAC Tape** | 3M or similar dead-soft aluminum tape (Shields plastic from UV-C radiation) | 1 | ₹350 | ₹350 | Amazon.in - Aluminum Tape |
| **PETG 3D Printer Filament** | 1KG Spool (Superior water/impact resistance compared to PLA) | 1 | ₹1,300 | ₹1,300 | Robu.in - eSUN PETG 1.75mm |
| **Brushed Aluminum Shell** | 60mm Outer Diameter Aluminum | 1 | ₹600 | ₹600 | *Local Pune Metal Fabrication /* |

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| | Pipe/Sleeve (Outer cosmetic chassis) | | | | *Hardware Market* |

## 6. The "Tiny Things" (Wiring & Micro-Components)

| Item | Description | Qty | Unit Cost (INR) | Total Cost (INR) | Buying Link / Source |
|---|---|---|---|---|---|
| **FR4 Double-Sided Perfboards** | Prototyping PCBs (Pack of 10) to mount the LM358 and custom circuits | 1 | ₹150 | ₹150 | Robu.in - FR4 PCB 5x7cm |
| **Kapton Tape** | High-temp polyimide tape to insulate stacked bare PCBs against short circuits | 1 | ₹150 | ₹150 | Robu.in - Kapton Tape 20mm |
| **AWG 24 Silicone Wire Kit** | Highly flexible, heat-resistant stranded wire (Red/Blk/Yel/Grn/Blu) | 1 | ₹450 | ₹450 | Robu.in - 24 AWG Wire Box |
| **Resistor & Capacitor Kit** | Box of assorted 1/4W resistors (10k, 4.7k etc.) and ceramic capacitors | 1 | ₹550 | ₹550 | Robu.in - Assorted R/C Box |
| **Heat Shrink Tubing Kit** | Assorted diameters for waterproofing wire splices | 1 | ₹200 | ₹200 | Robu.in - Heat Shrink Box |
| **LM358 Dual Op-Amp IC** | Used for conditioning the analog Piezoelectric acoustic signals | 5 | ₹10 | ₹50 | Robu.in - LM358 |

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| **Soldering Wire & Wick** | High-quality 60/40 rosin core solder and copper desoldering wick | 1 | ₹300 | ₹300 | Robu.in - Solder & Wick Kit |

**GRAND TOTAL FOR APPROVAL: ₹24,478.00 INR**